Quiz 1 - DSC 10, Fall 2023

Full Name:

PID:

Discussion: ◯ A (3PM) ◯ B (4PM) ◯ C (5PM)

Instructions:

- This quiz consists of 4 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles; if we cannot tell which option(s) you selected, you may lose points. For multiple choice questions, select **only one answer**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- No aids are allowed (no notes, no calculators, and no computers).

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

Question 1

After a trip to the zoo, Anthony wrote the following block of code.

```
zebra = 5
lion = 4
cow = 1
zebra = zebra * 2
lion = abs(cow - lion)
zebra = zebra + lion ** 2
cow = (zebra + lion) / 2 * lion
```

After running the above block of code, what is the value of `cow`?

`cow`

Question 2

Consider the following four assignment statements.

```
bass = "5"
tuna = 2
sword = ["4.0", 5, 12.5, -10, "2023"]
gold = [4, "6", "CSE", "doc"]
```

a) What is the value of the expression `bass * tuna`?

`bass * tuna`

b) Which of the following expressions results in an error?

◯ `int(sword[0])` ◯ `float(sword[1])` ◯ `int(sword[2])`
◯ `int(sword[3])` ◯ `float(sword[4])`

c) Which of the following expressions evaluates to `"DSC10"`?

◯ `gold[3].replace("o", "s").title() + str(gold[0] + gold[1])`
◯ `gold[3].replace("o", "s").upper() + str(gold[0] + int(gold[1]))`
◯ `gold[3].replace("o", "s").upper() + str(gold[1] + int(gold[0]))`
◯ `gold[3].replace("o", "s").title() + str(gold[0] + int(gold[1]))`

PID: _____

Question 3

Consider the following assignment statement.

```
puffin = np.array([5, 9, 13, 17, 21])
```

a) Provide arguments to call `np.arange` with so that the array `penguin` is identical to the array `puffin`.

`penguin = np.arange(` [ ] `)`

b) Fill in the blanks so that the array `parrot` is also identical to the array `puffin`. *Hint: Start by choosing* `y` *so that* `parrot` *has length 5.*

```
parrot = __(x)__ * np.arange(0, __(y)__, 2) + __(z)__
```

x: [ ] , y: [ ] , z: [ ]

Question 4

Suppose `students` is a DataFrame of all students who took DSC 10 last quarter. `students` has one row per student, where:

- The index contains students' PIDs as strings starting with `"A"`.
- The `"Overall"` column contains students' overall percentage grades as floats.
- The `"Animal"` column contains students' favorite animals as strings.

a) What type is `students.get("Overall")`? If this expression errors, select "this errors."

◯ float ◯ string ◯ array ◯ Series ◯ this errors

b) What type is `students.get("PID")`? If this expression errors, select "this errors."

◯ float ◯ string ◯ array ◯ Series ◯ this errors

Vanessa is one student who took DSC 10 last quarter. Her PID is A12345678, she earned the sixth-highest overall percentage grade in the class, and her favorite animal is the giraffe.

c) Supposing that `students` is already sorted by `"Overall"` in **descending** order, fill in the blanks so that `animal_one` and `animal_two` **both** evaluate to `"giraffe"`.

```
animal_one = students.get(__(x)__).loc[__(y)__]
animal_two = students.get(__(x)__).iloc[__(z)__]
```

x: [ ] , y: [ ] , z: [ ]

d) If `students` wasn't already sorted by `"Overall"` in descending order, which of your answers would need to change?

◯ Neither `y` nor `z` would need to change ◯ `y` only
◯ Both `y` and `z` would need to change ◯ `z` only

Before submitting your quiz, make sure your PID is on the front page and on the top right corner of page 3.